Topological quantum computation: Majorana fermions in hybrid superconductor-semiconductor nanowire devices

EE451 - Supervised Research Exposition presentation

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Motive -> Current implementations of quantum computers suffer greatly from decoherence of qubits

Aim in mind -> to contribute in devising a way to perform fault-tolerant quantum computation

Topologically-protected quantum computation: braiding

<- A caricature

Topologically-protected quantum computation: braiding

Quantum knot:

-> a topologically protected entity

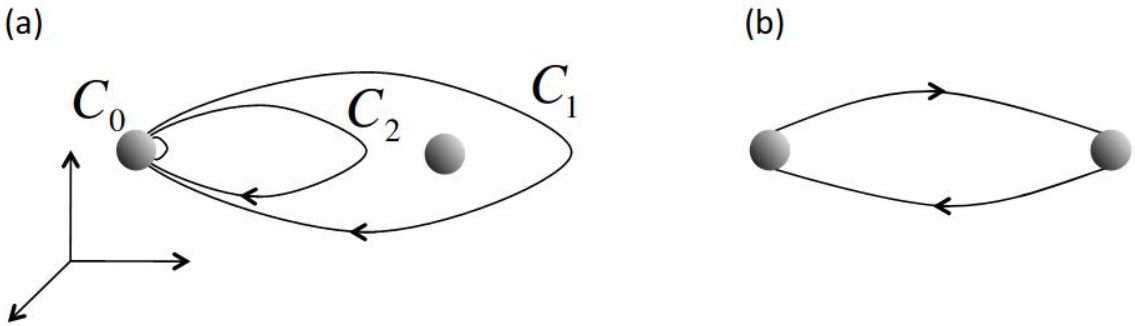
-> exists as braids in the world lines of exotic particles non-Abelian anyons

-> under smooth deformations of world lines, mathematical description of braid remains unchanged, protecting the knot from local perturbations in the world lines of the non-Abelian anyons.*

<- A caricature
Braided world lines remain topologically equivalent upto smooth deformations.

*Computing with quantum knots - Scientific American, 2006

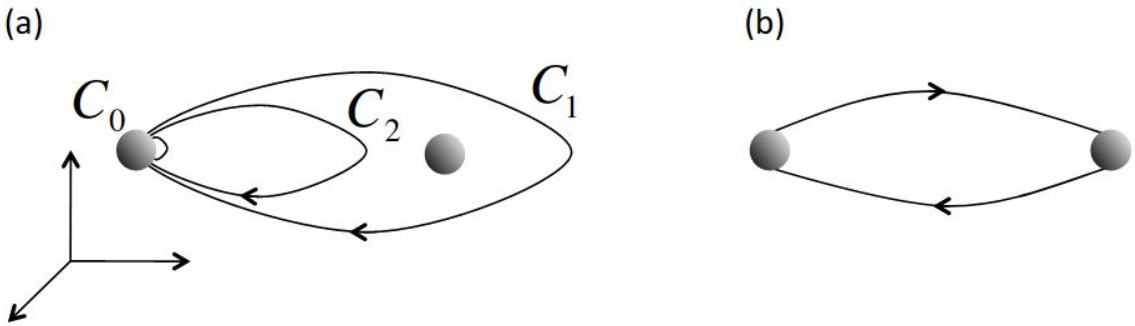
Non-Abelian anyonic statistics



(a) A particle spans a loop around another one. In three dimensions it is possible to continuously deform the path C_1 to the path C_2, which is equivalent to a trivial path, C_0. (b) Two successive exchanges between two particles are equivalent to a circulation of one particle around the other and a translation.

In 3+1 dimensions -> exchange statistics of bosons and fermions

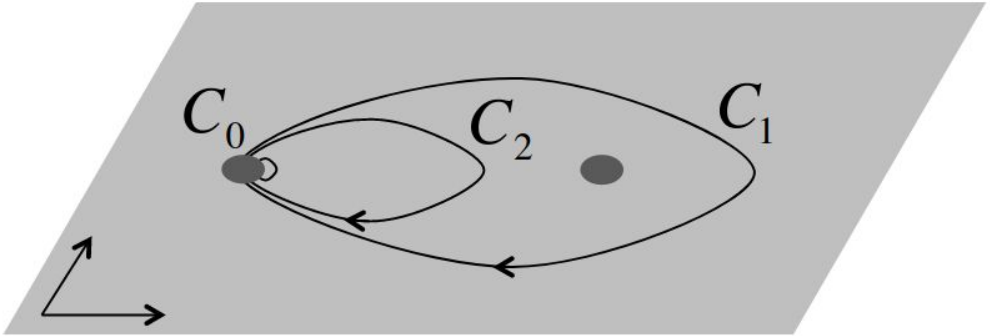
Non-Abelian anyonic statistics



(a) A particle spans a loop around another one. In three dimensions it is possible to continuously deform the path C_1 to the path C_2, which is equivalent to a trivial path, C_0. (b) Two successive exchanges between two particles are equivalent to a circulation of one particle around the other and a translation.

In 3+1 dimensions -> exchange statistics of bosons and fermions
In 2+1 dimensions -> exotic anyonic exchange statistics also possible

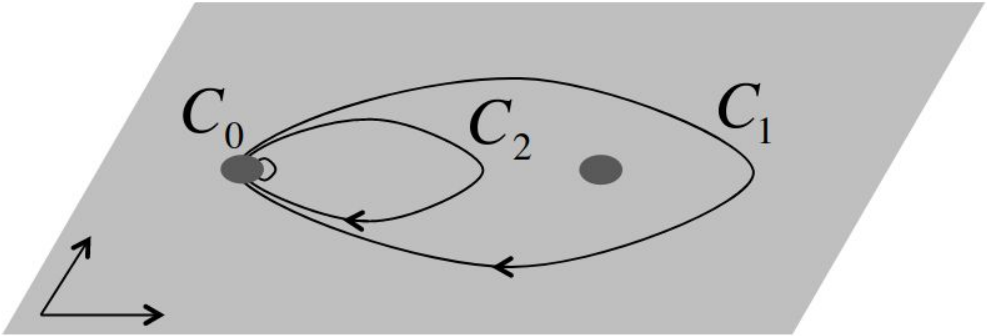
Non-Abelian anyonic statistics



In two dimensions the two paths C_1 and C_2 are topologically distinct. This gives the possibility of having non-trivial phase factors appearing when one particle circulates around the other. In other words, one can assign a non-trivial unitary to the evolution corresponding to path C_1.

Unitary because -> **adiabatic** evolution of wavefunctions is unitary

Topological quantum computation: Non-Abelian anyonic statistics



Trivial unitary evolution -> $\psi(\mathbf{r}_1, \mathbf{r}_2) \longrightarrow e^{i\theta}\psi(\mathbf{r}_2, \mathbf{r}_1)$

$$\psi(\mathbf{r}_1, \mathbf{r}_2) \longrightarrow e^{i2\theta}\psi(\mathbf{r}_1, \mathbf{r}_2)$$

The special cases, θ = 0, π correspond to bosons and fermions, respectively, while any other value corresponds to **Abelian anyons**

Non-trivial unitary evolution -> $\psi_a \longrightarrow U_{ab}\psi_b$ -> corresponds **to non-Abelian anyons**

Reference - [1], [2]

Simplest way of getting non-Abelian statistics -> Majorana fermions!

Non-Abelian braiding statistics and the Majorana fermion exchange operator

Majorana fermions, would reproduce non-Abelian braiding statistics!

$$\sigma_i \sigma_j = \sigma_j \sigma_i, |i-j| \geq 2$$

$$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}, 1 \leq i \leq n-1$$

If we set -> $\sigma_k = (1 + \gamma_{k+1}\gamma_k)/\sqrt{2}$

Here γ_{k+1} and γ_k are the (k+1)th and the kth Majorana operators respectively, defined as -

$$\{\gamma_i^A, \gamma_j^B\} = 2\delta_{ij}\delta_{AB}\,, \quad \gamma_j = \gamma_j^\dagger\,, \quad \gamma_j^2 = \gamma_j^{\dagger 2} = 1$$

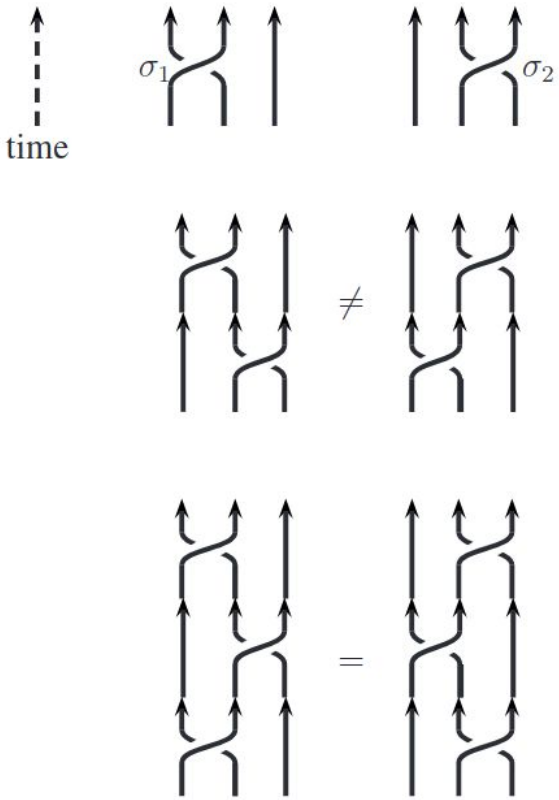
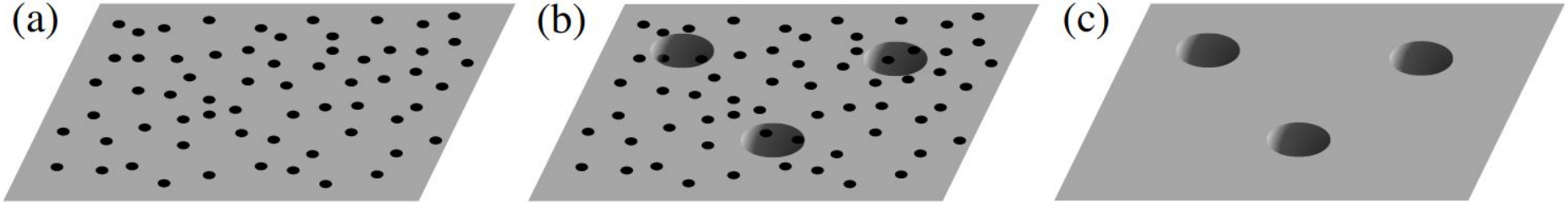


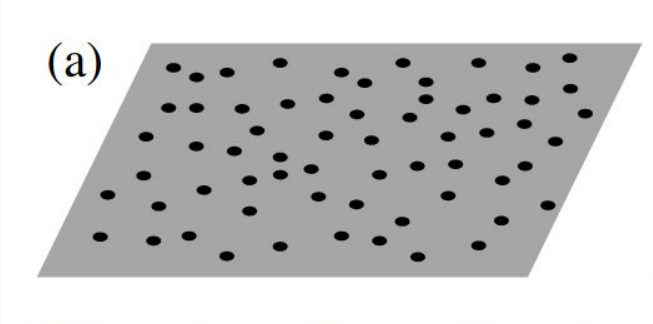
FIG. 1. Graphical representation of elements of the braid group. Top: The two elementary braid operations σ_1 and σ_2 on three particles. Middle: Shown here $\sigma_2\sigma_1 \neq \sigma_1\sigma_2$; hence the braid group is non-Abelian. Bottom: The braid relation [Eq. (3)] $\sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1}$.

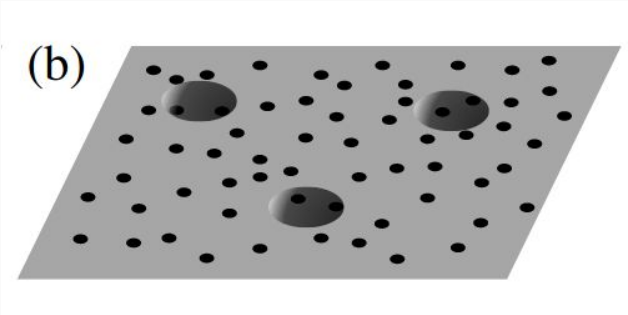
References - [2], [3]

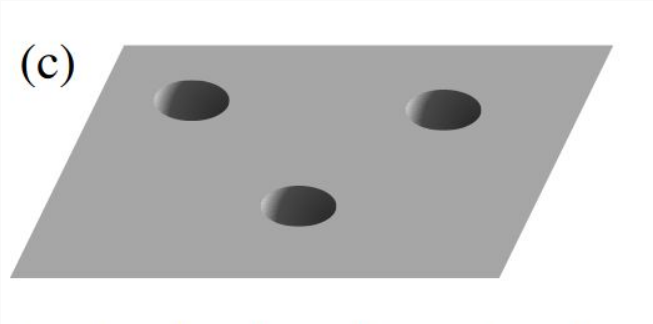
Getting Non-Abelian anyons as collective excitations:



(a) A system with constituent particles confined on a plane that give rise to a two-dimensional wave function. (b) Quasiparticles are identified as localised properties of the two-dimensional wave function of the constituent particles. (c) Often we forget the constituent particles and we treat the quasiparticles as elementary ones living on the two-dimensional space.


(a)


(b)


(c)

Getting Non-Abelian anyons as collective excitations

Kitaev's proposal (Kitaev chain) -
Engineering Majorana bound states in a p-wave superconductor

$$H = -\mu \sum_{j=1}^{N} \left(c_j^\dagger c_j - \frac{1}{2} \right) + \sum_{j=1}^{N-1} \left[-t \left(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j \right) + \Delta\, c_j c_{j+1} + \Delta^*\, c_{j+1}^\dagger c_j^\dagger \right]$$

Lattice with N fermionic sites - each of which can be empty or filled

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Considering the definition: $\gamma_j^A = c_j + c_j^\dagger \,, \quad \gamma_j^B = i(c^\dagger - c_j)$

From the fermionic anti-commutation relations -

$$\{c_i^\dagger, c_j^\dagger\} = \{c_i, c_j\} = 0 \,, \quad \{c_i^\dagger, c_j\} = c_i^\dagger c_j + c_j c_i^\dagger = \delta_{ij}$$

we get Majorana fermion operators!

$$\{\gamma_i^A, \gamma_j^B\} = 2\delta_{ij}\delta_{AB} \,, \quad \gamma_j = \gamma_j^\dagger \,, \quad \gamma_j^2 = \gamma_j^{\dagger 2} = 1$$

Getting Non-Abelian anyons as collective excitations

Kitaev's proposal (Kitaev chain) -
The Hamiltonian

$$H = -\mu \sum_{j=1}^{N} \left(c_j^\dagger c_j - \frac{1}{2} \right) + \sum_{j=1}^{N-1} \left[-t \left(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j \right) + \Delta\, c_j c_{j+1} + \Delta^*\, c_{j+1}^\dagger c_j^\dagger \right]$$

then becomes

$$H = -\frac{i\mu}{2} \sum_{j=1}^{N} \gamma_j^A \gamma_j^B + \frac{i}{2} \sum_{j=1}^{N-1} \left[(\Delta + t) \gamma_j^B \gamma_{j+1}^A + (\Delta - t) \gamma_j^A \gamma_{j+1}^B \right]$$

Getting Non-Abelian anyons as collective excitations

Kitaev's proposal (Kitaev chain) -

$$H = -\frac{i\mu}{2}\sum_{j=1}^{N}\gamma_j^A\gamma_j^B + \frac{i}{2}\sum_{j=1}^{N-1}\left[(\Delta+t)\gamma_j^B\gamma_{j+1}^A + (\Delta-t)\gamma_j^A\gamma_{j+1}^B\right]$$

There are two cases possible -

The topologically trivial case - $t = \Delta = 0 \longrightarrow H = -\frac{i\mu}{2}\sum_{j=1}^{N}\gamma_j^A\gamma_j^B$

The topologically non-trivial case -

$$t = \Delta \neq 0, \mu = 0 \longrightarrow H = it\sum_{j=1}^{N-1}\gamma_j^B\gamma_{j+1}^A = i\Delta\left[\gamma_1^B\gamma_2^A + \gamma_2^B\gamma_3^A + \cdots + \gamma_{N-1}^B\gamma_N^A\right]$$

References - [4]

Getting Non-Abelian anyons as collective excitations

Kitaev's proposal (Kitaev chain) -

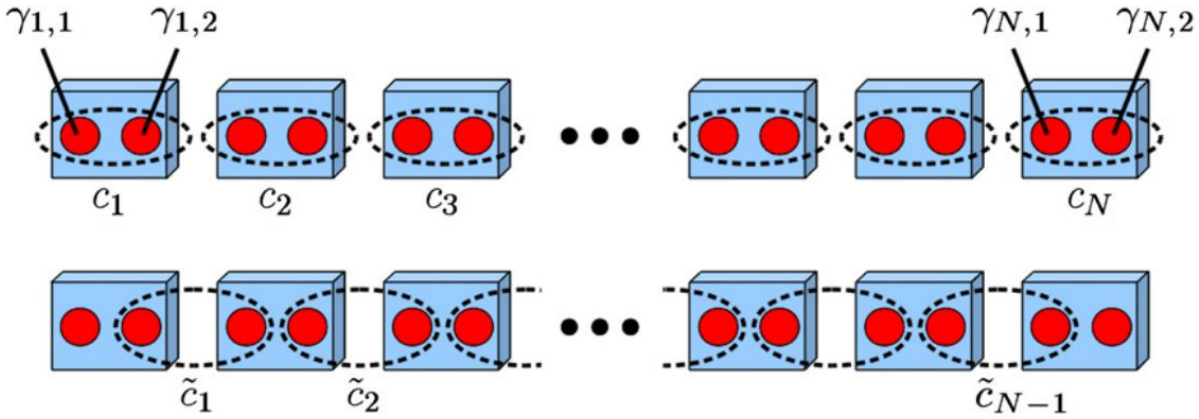

Figure 1. Sketch of Kitaev's 1D p-wave superconducting tight binding chain. Upper panel: the fermion operators on each site i of the chain can be split into two Majorana operators, $\gamma_{i,1}$ and $\gamma_{i,2}$. Lower panel: in the limit $\mu = 0$, $t = \Delta$, the Hamiltonian is diagonal in fermion operators which are obtained by combining instead Majorana operators on neighboring sites, $\gamma_{i+1,1}$ and $\gamma_{i,2}$. This leaves two unpaired Majorana operators, $\gamma_{1,2}$ and $\gamma_{N,1}$, which can be combined to form one zero energy, highly non-local fermion operator, $\tilde{c}_M$.

Getting Non-Abelian anyons as collective excitations

Kitaev's proposal (Kitaev chain) -

$$H = it\sum_{j=1}^{N-1}\gamma_j^B\gamma_{j+1}^A = i\Delta\left[\gamma_1^B\gamma_2^A + \gamma_2^B\gamma_3^A + \cdots + \gamma_{N-1}^B\gamma_N^A\right]$$

Considering the definition: $d_j = \frac{1}{2}\left(\gamma_j^B + i\gamma_{j+1}^A\right), \quad d_j^\dagger = \frac{1}{2}\left(\gamma_j^B - i\gamma_{j+1}^A\right)$

$$H = 2t\sum_{j=1}^{N-1}\left(d_j^\dagger d_j - \frac{1}{2}\right)$$

The remaining Majorana pair form a fermion - $f = \frac{1}{2}\left(\gamma_1^A + i\gamma_N^B\right), \quad f^\dagger = \frac{1}{2}\left(\gamma_1^A - i\gamma_N^B\right)$

Which has zero energy -> Majorana 'zero mode'

References - [4]

$$d_j = \frac{1}{2}\left(\gamma_j^B + i\gamma_{j+1}^A\right),$$

$$d_j^\dagger = \frac{1}{2}\left(\gamma_j^B - i\gamma_{j+1}^A\right)$$

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Getting Non-Abelian anyons as collective excitations

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$$H = -\mu \sum_{j=1}^{N} \left(c_j^\dagger c_j - \frac{1}{2} \right) + \sum_{j=1}^{N-1} \left[-t \left(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j \right) + \Delta\, c_j c_{j+1} + \Delta^*\, c_{j+1}^\dagger c_j^\dagger \right]$$

"Now, in order to investigate the properties of the superconducting bulk without perturbations from the ends of the chain and to fully describe the two phases we have studied previously, we assume that the chain forms a closed loop with periodic boundary conditions." [4]

$$H = -\mu \sum_{j=1}^{N} \left(c_j^\dagger c_j - \frac{1}{2} \right) + \sum_{j=1}^{N} \left[-t \left(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j \right) + \Delta\, c_j c_{j+1} + \Delta^*\, c_{j+1}^\dagger c_j^\dagger \right]$$

Getting Non-Abelian anyons as collective excitations

Kitaev's proposal (Kitaev chain) -

Sites 1 and N+1 correspond to the same site; considering the Hamiltonian in momentum space

$$H = \frac{1}{2}\sum_k \psi_k^\dagger H_{BdG}\psi_k + \frac{1}{2}\sum_k(-2t\cos ka)\,, \quad \psi_k = \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix}$$

$$H_{BdG} = \xi_k\tau_z + \Delta_k\tau_y = \mathbf{h}\cdot\tau$$

$$\mathbf{h} = (0, \Delta_k, \xi_k),\ \Delta_k = -2\Delta\sin ka,\ \xi_k = -\mu - 2t\cos ka,\ \text{and}\ \tau = (\tau_x, \tau_y, \tau_z)$$

Energy spectrum of H_{BdG} is given by $E_{k,\pm} = \pm\sqrt{(\mu + 2t\cos ka)^2 + 4\Delta^2\sin^2 ka}$

References - [4]

Getting Non-Abelian anyons as collective excitations

Kitaev's proposal (Kitaev chain) -

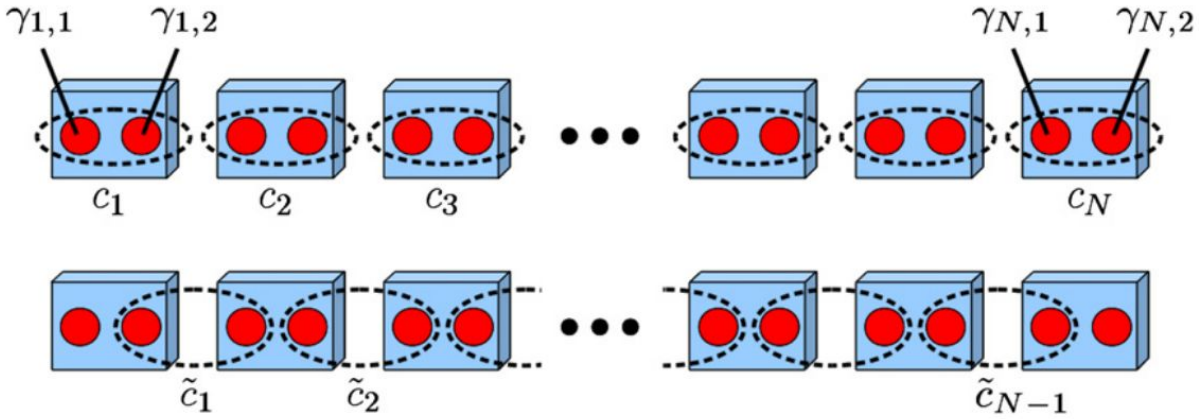
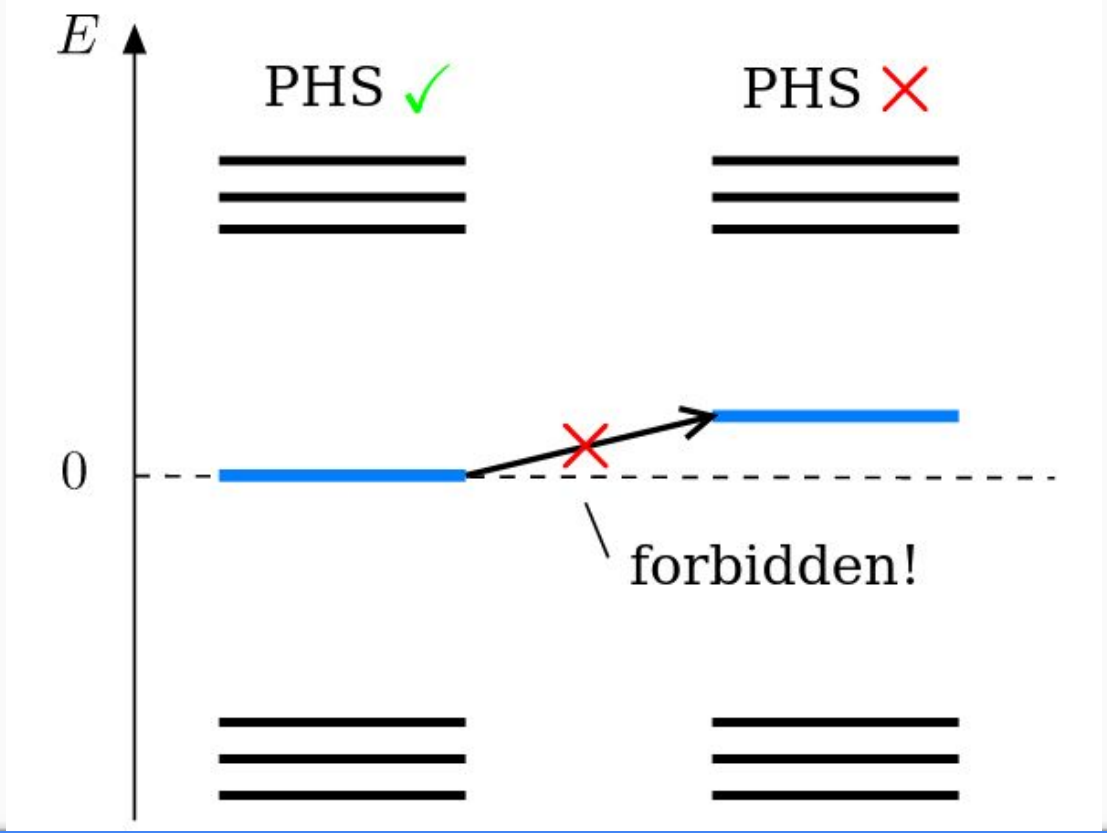


Figure 1. Sketch of Kitaev's 1D p-wave superconducting tight binding chain. Upper panel: the fermion operators on each site i of the chain can be split into two Majorana operators, $\gamma_{i,1}$ and $\gamma_{i,2}$. Lower panel: in the limit $\mu = 0$, $t = \Delta$, the Hamiltonian is diagonal in fermion operators which are obtained by combining instead Majorana operators on neighboring sites, $\gamma_{i+1,1}$ and $\gamma_{i,2}$. This leaves two unpaired Majorana operators, $\gamma_{1,2}$ and $\gamma_{N,1}$, which can be combined to form one zero energy, highly non-local fermion operator, $\tilde{c}_M$.

Roots of topological protection of Majorana edge modes



References - [6]

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Energy spectrum of H_{BdG} is given by $E_{k,\pm} = \pm\sqrt{(\mu + 2t\cos ka)^2 + 4\Delta^2\sin^2 ka}$

References - [4]

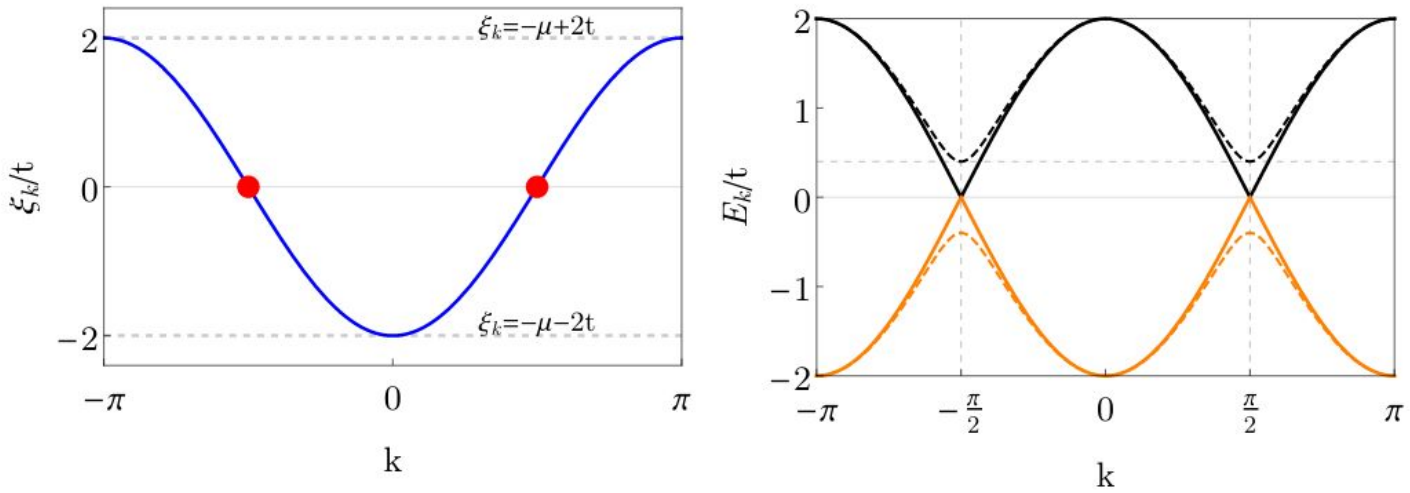

FIGURE 1.6: (Color online) (left) Energy dispersion of the normal system in the Kitaev's model $\xi_k = -\mu - 2t\cos ka$ with $a = 1$. The two horizontal dashed lines represent a closure of the energy gap when the chemical potential is tuned to $\mu = -2t$ for $k = 0$ and $\mu = 2t$ for $k = \pm\pi$. Within the region $-2t < \mu < 2t$ one finds an odd number of pairs of Fermi points (red filled circles) for $-\pi < k < \pi$, where the topological invariant $\nu = -1$ leads to a clear signature of the topological superconducting phase with Majorana zero modes located at the ends of the chain. Outside of such region, the system is in the non-topological phase. (right) Quasiparticle excitation spectrum given by Eq. (1.35) with $\mu = 0$, where $\Delta = 0$ (solid curves) and $\Delta = 0.2t$ (dashed curves). Notice that at $k = \pm\pi/2$ the superconducting pairing opens a gap 2Δ (dashed horizontal line).

References - [4]

Physical implementation -
Though there is ongoing research effort, we haven’t had any good known p-wave superconductors so far.
What to do now?

Make a system with a Hamiltonian which resembles that of the p-wave superconductor

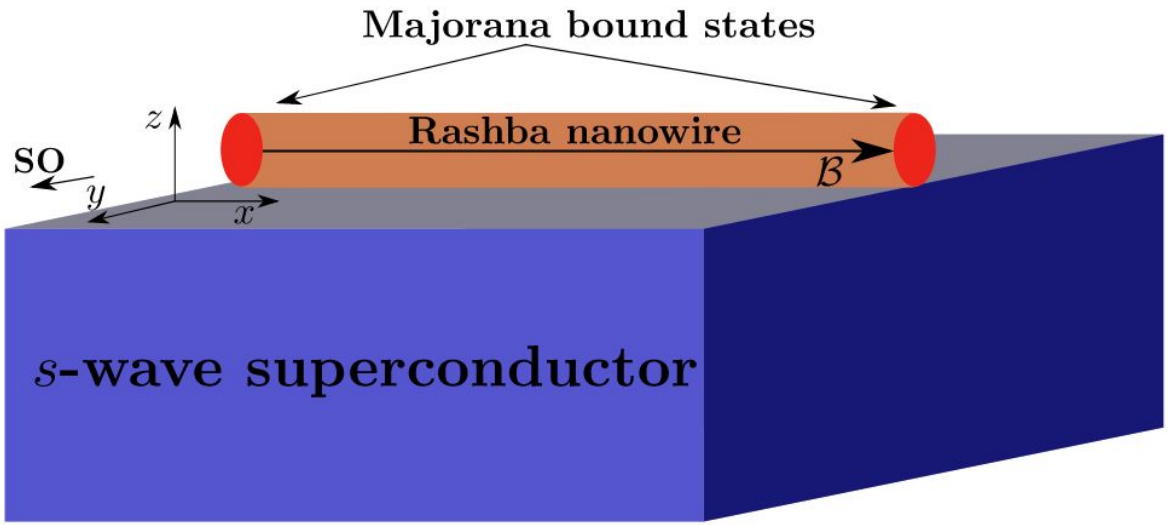

FIGURE 1.7: (Color online) Sketch of a 1D semiconducting Rashba nanowire placed on a s-wave superconductor. A magnetic field $\mathcal{B}$ is applied along the wire x-axis, which is perpendicular to the spin-orbit field. These ingredients represent a solid platform for investigating the topological superconducting phase and therefore Majorana bound states, red filled circles, in 1D condensed matter systems.

References - [4]

Nanowire with Rashba spin-orbit coupling

$$H_0 = \frac{\hbar^2 k_x^2}{2m} - \mu - \alpha_{SO} k_x \sigma_y + E_Z \sigma_z$$

σ_x

$$E_\pm = \frac{\hbar^2 k_x^2}{2m} - \mu \pm \sqrt{\alpha_{SO}^2 k_x^2 + E_Z^2}$$

Nanowire with Rashba spin-orbit coupling

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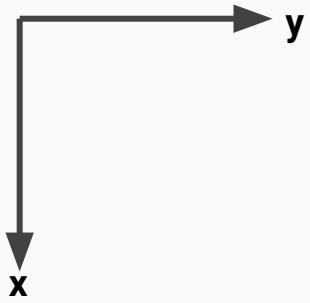
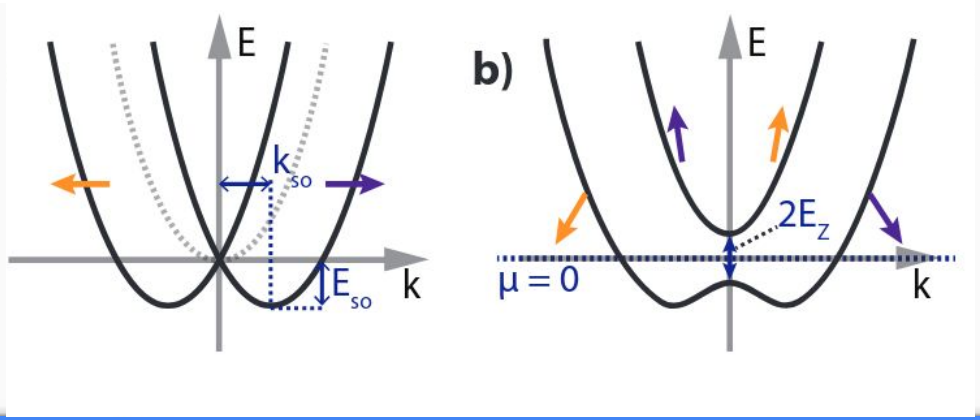
$$E_\pm = \frac{\hbar^2 k_x^2}{2m} - \mu \pm \sqrt{\alpha_{SO}^2 k_x^2 + E_Z^2}$$

$$\mathbf{B}_{SO} = B_{SO}\hat{\mathbf{y}} = \frac{2\alpha_{SO}}{\hbar} k_x \hat{\mathbf{y}}$$



Spin orientation as in plots

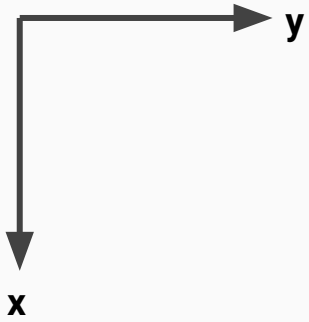


Nanowire with Rashba spin-orbit coupling with proximity effect

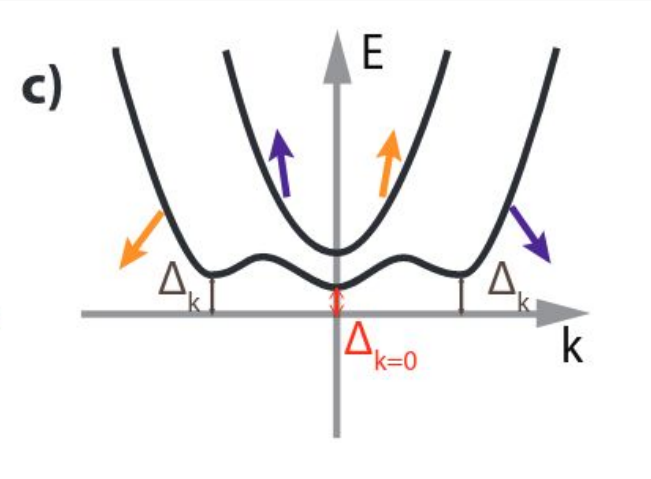
$$H_{\text{eff}} = \int_{-\infty}^{\infty} \left(\Psi_x^\dagger \left(\frac{\hbar^2 k_x^2}{2m} - \mu - E_Z \right) \Psi_x + e^{i\phi} \frac{\alpha_{SO} \hbar k_x}{2E_Z} \Delta \Psi_x \Psi_x + \text{H.c.} \right) \mathrm{d}x \qquad (2.43)$$

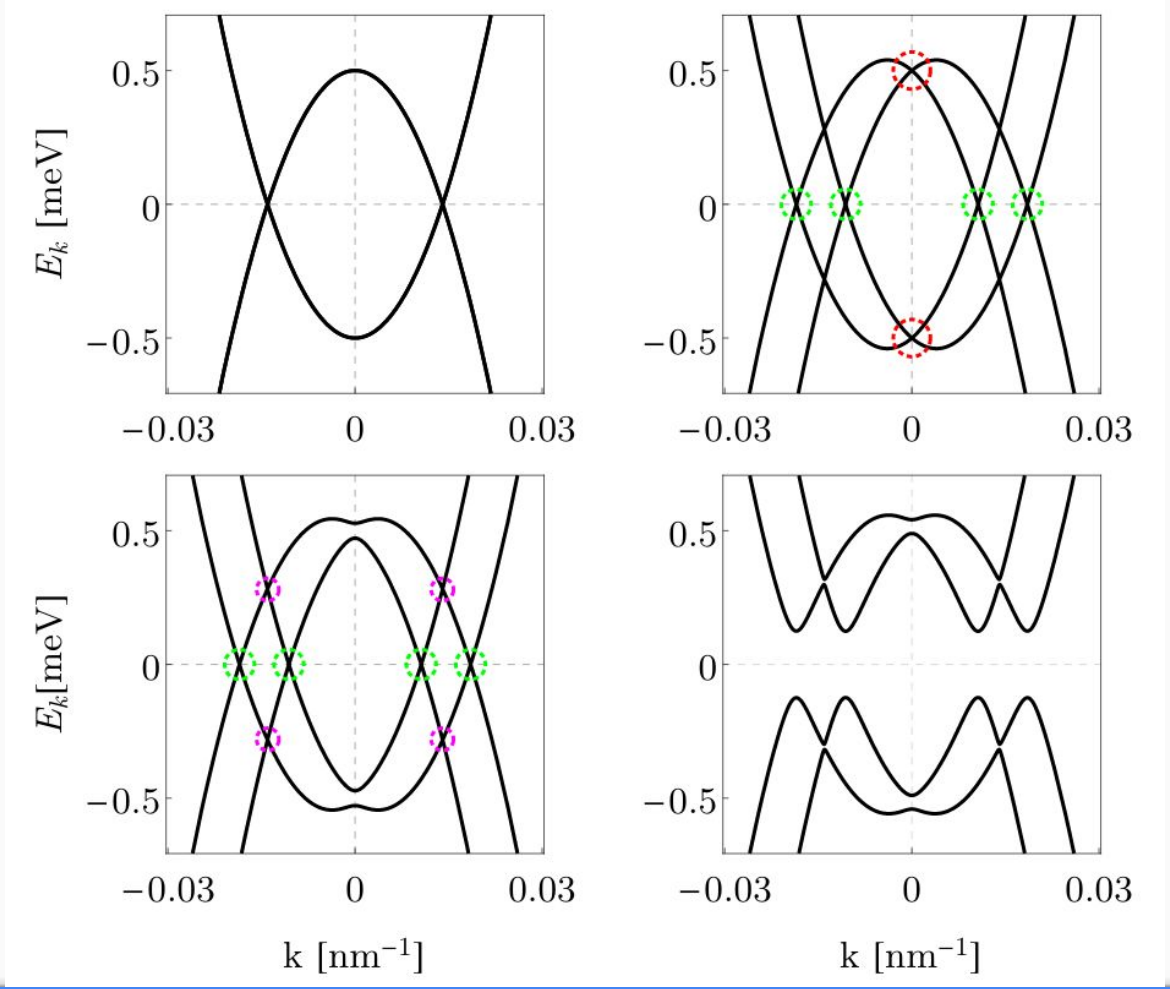
Introducing an effective chemical potential $\mu_{\text{eff}} = \mu - E_Z$ and an effective superconducting paring strength $\Delta_{\text{eff}} = (\alpha_{SO}\Delta) / (2E_Z)$ shows that this Hamiltonian is exactly the same as that of an 1D p-wave superconductor as given in (2.34) and the system constructed is a realization of the Kitaev chain introduced before.

(2.34) corresponds to the continuum representation of the Kitaev chain Hamiltonian discussed before



Spin orientation as in plots

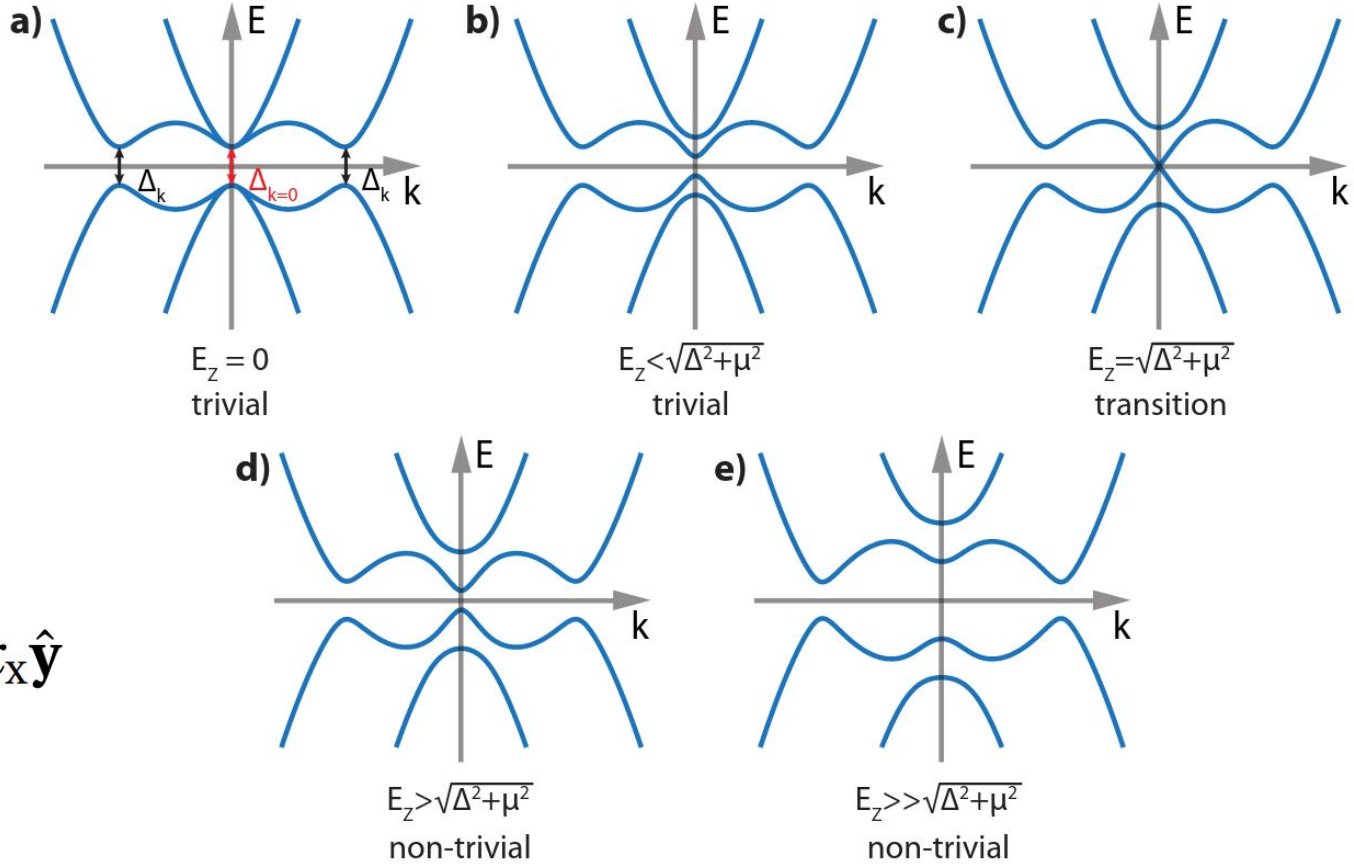




References - [4]

$$\mu = 0$$

$$\mathbf{B}_{\mathrm{SO}} = B_{\mathrm{SO}}\hat{\mathbf{y}} = \frac{2\alpha_{\mathrm{SO}}}{\hbar} k_{\mathrm{x}}\hat{\mathbf{y}}$$



$$E_{\mathrm{gap,k=0}} = \sqrt{\Delta^2 + \mu^2} - E_{\mathrm{Z}}$$

References - [7]

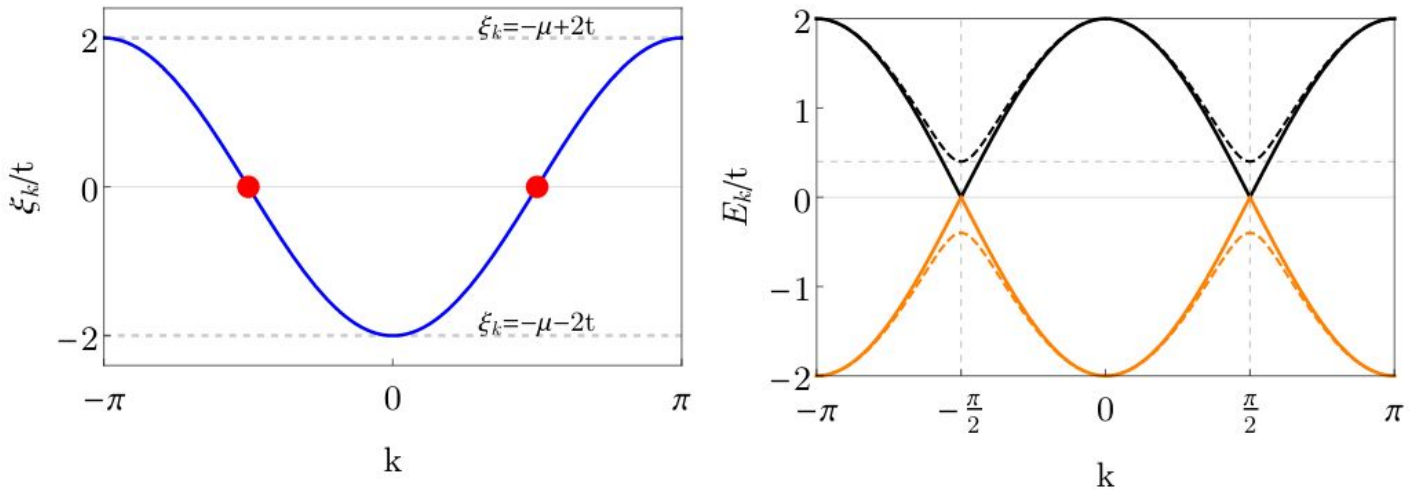


FIGURE 1.6: (Color online) (left) Energy dispersion of the normal system in the Kitaev's model $\xi_k = -\mu - 2t\cos ka$ with $a = 1$. The two horizontal dashed lines represent a closure of the energy gap when the chemical potential is tuned to $\mu = -2t$ for $k = 0$ and $\mu = 2t$ for $k = \pm\pi$. Within the region $-2t < \mu < 2t$ one finds an odd number of pairs of Fermi points (red filled circles) for $-\pi < k < \pi$, where the topological invariant $\nu = -1$ leads to a clear signature of the topological superconducting phase with Majorana zero modes located at the ends of the chain. Outside of such region, the system is in the non-topological phase. (right) Quasiparticle excitation spectrum given by Eq. (1.35) with $\mu = 0$, where $\Delta = 0$ (solid curves) and $\Delta = 0.2t$ (dashed curves). Notice that at $k = \pm\pi/2$ the superconducting pairing opens a gap 2Δ (dashed horizontal line).

References - [4]

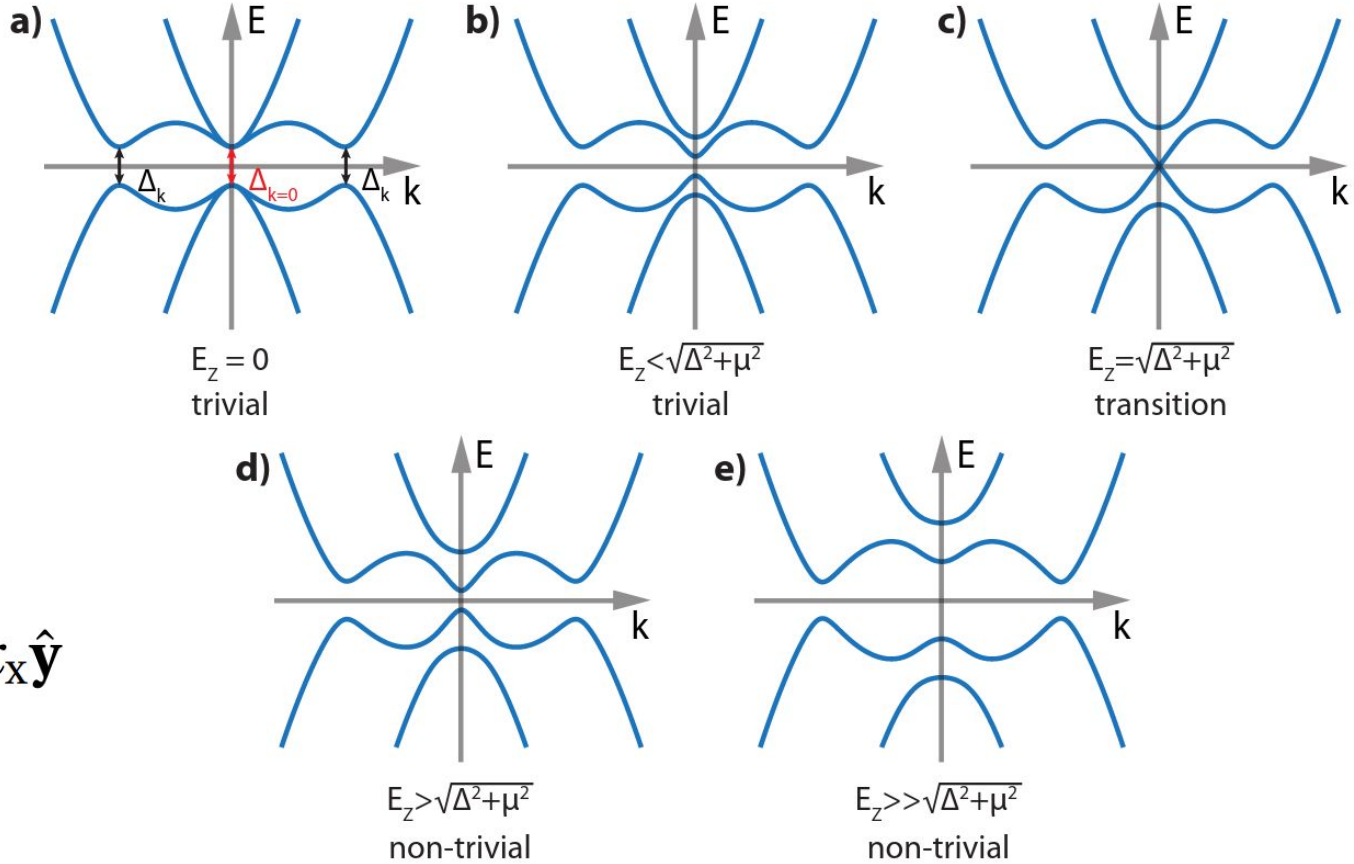
Effective superconducting gap is determined by the minimum of the two gaps

$$E_{\mathrm{gap},k=k_f} = \frac{2\Delta E_{SO}}{\left(E_{SO}(2E_{SO} + \sqrt{E_Z^2 + 4E_{SO}^2})\right)^{1/2}}$$

References - [7]

$$\mu = 0$$

$$\mathbf{B}_{\mathrm{SO}} = B_{\mathrm{SO}}\hat{\mathbf{y}} = \frac{2\alpha_{\mathrm{SO}}}{\hbar} k_{\mathrm{x}}\hat{\mathbf{y}}$$



$$E_{\mathrm{gap,k=0}} = \sqrt{\Delta^2 + \mu^2} - E_{\mathrm{Z}}$$

References - [7]

Effective superconducting gap is determined by the minimum of the two gaps

$$E_{\mathrm{gap,k=k_f}} = \frac{2\Delta E_{\mathrm{SO}}}{\left(E_{\mathrm{SO}}(2E_{\mathrm{SO}} + \sqrt{E_{\mathrm{Z}}^2 + 4E_{\mathrm{SO}}^2})\right)^{1/2}}$$

$$E_z >> E_{so}, E_{gap,k=k_f} = \frac{2\Delta\sqrt{E_{so}}}{\sqrt{E_z}}$$

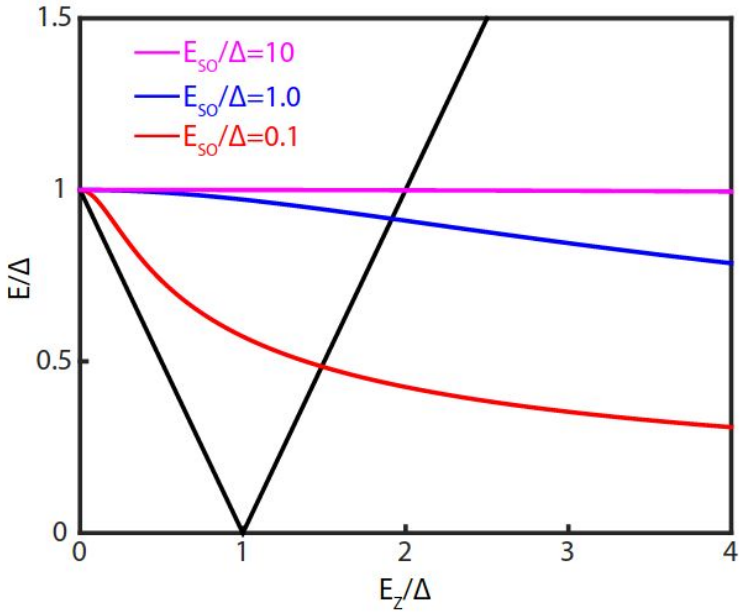


Figure 2.4 |Spectral gaps as a function of magnetic field in a superconducting 1D nanowire with Rashba SOI. Gaps are shown for $\mu = 0$. Black line is the gap at $k = 0$ which is indepent of SOI strength. Colored lines represent the gap at finite k for different SOI strength. The gap at $k = 0$ closes and reopens linearly determining the effective gap at low B. Between $E_Z/\Delta = 1$ and $E_Z/\Delta = 2$, the finite k gap becomes smaller as the $k = 0$ gap, thus determining the effective gap at higher B fields.

References - [7]

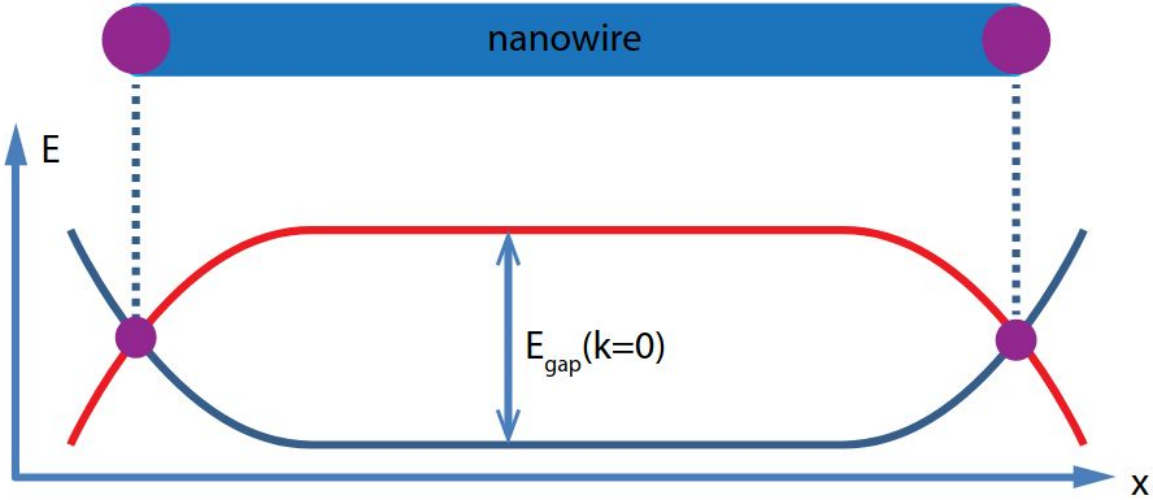
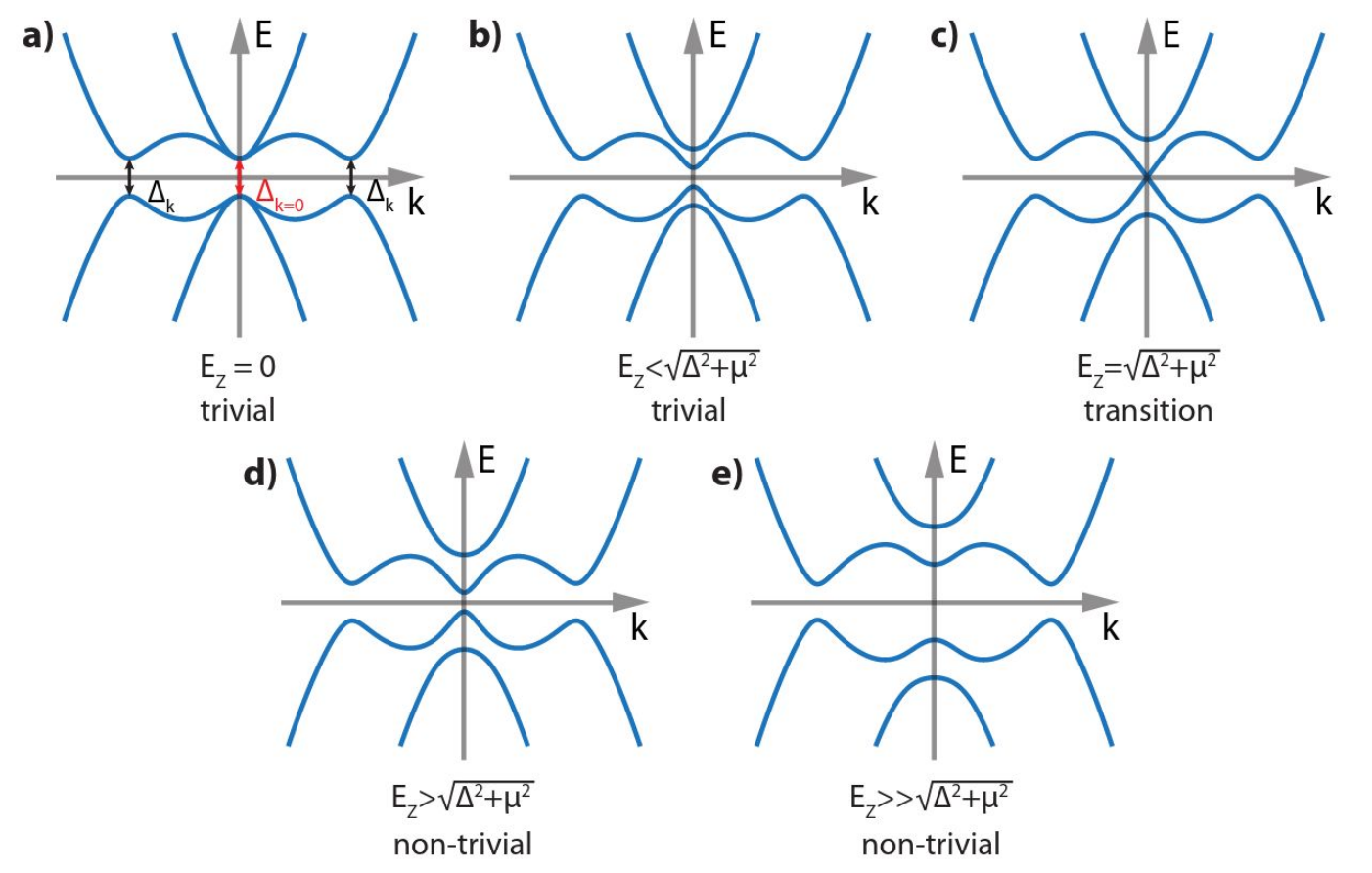


Figure 2.5 |Schematic of MBS arising at the ends of the 1D nanowire. Once the nanowire has undergone the topological phase transition and the gap is inverted, it as to close at the wire ends to connect the system to the topologically trivial vacuum. The case depicted here corresponds to μ diverging to $-\infty$ at the wire ends. The point of gap closure corresponds to a localized, single MBS. This immediately implies that the MBS's come in pairs, at both ends of a topologically non-trivial section a MBS is present.

References - [7]

$\mu = 0$



$$E_{\text{gap},k=0} = \sqrt{\Delta^2 + \mu^2} - E_Z$$

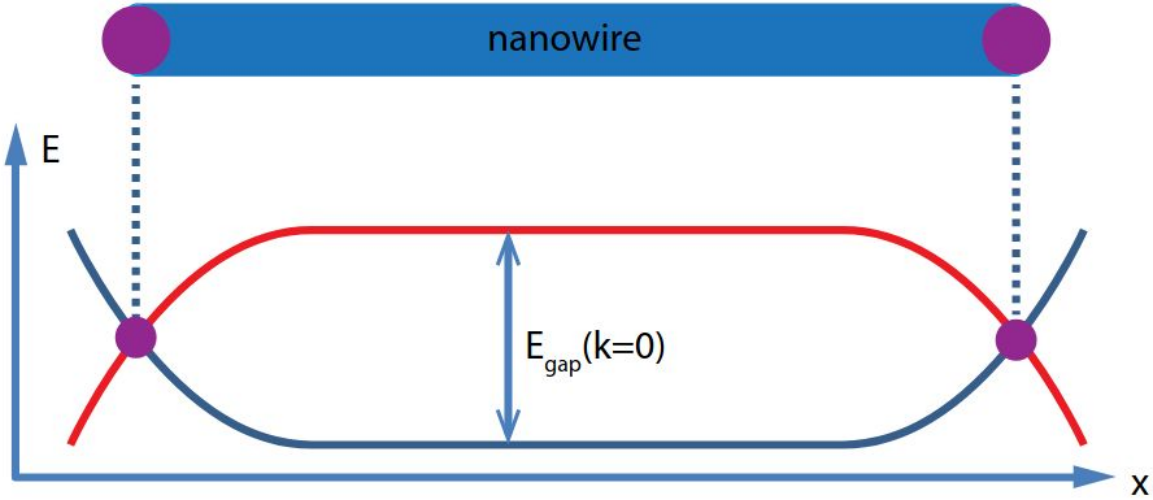


Figure 2.5 |Schematic of MBS arising at the ends of the 1D nanowire. Once the nanowire has undergone the topological phase transition and the gap is inverted, it as to close at the wire ends to connect the system to the topologically trivial vacuum. The case depicted here corresponds to μ diverging to $-\infty$ at the wire ends. The point of gap closure corresponds to a localized, single MBS. This immediately implies that the MBS's come in pairs, at both ends of a topologically non-trivial section a MBS is present.

References - [7]

References

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[7] Signatures of Majorana Fermions in Hybrid Superconductor-Semiconductor Nanowire Devices (Doctoral thesis, 2016 Zuo, K. Mourik, V.) https://doi.org/10.4233/uuid:3fd0aed5-000c-4e50-a772-269436637fc4